Students' Performance Prediction Using Multimodal Machine Learning

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***Abstract*—Technological intervention in the field of education has gained significant relevance, especially during the post-pandemic era. The three dimensions of interaction that influence learning are the student's interaction with the content, peers, and instructors. Learning ecosystems are expected to ensure these interactions in a seamless way. Technological interventions have provided us with provisions to establish the interactions. The data that we obtain while the student is interacting with content, peers, and instructors can serve as feedback to students and instructors. The motivation of the current study lies in the direction of investigating 'what' and 'how' current practices of establishing the interaction with content, peers, and instructor are influencing students' performance. The other dimension includes how demographic factors like gender influence the performance of students when technological interventions are made.The sample considered in the study included 140 first-year engineering students in a private university. The outcome of the study helped to do early prediction of student failures and identification of factors that influences the student's success. The data for the study was collected from multiple modalities. Clickstream data was collected from a learning management system to understand the interaction of students with the course content. Student collaboration data was collected from GitHub to understand the interaction of students with peers. Demographic data was collected from student academic performance to understand how past performance and demographic factors influence future performance.The findings reveal that the student interaction with the content and the student performance have a positive relationship with a correlation coefficient of 0.68. The algorithms including random forest, naive Bayes, decision tree, support-vector Machine, and extreme gradient boosting were used to perform multiclass classification to predict the performance. The students were grouped into four classes including 'Excellent', 'Good', 'Average', and 'Poor' using decision tree with a classification accuracy of 96%.**

Keywords*—*Performance prediction, Learning Ecosystems, Student Interactions, Demographic Factors

I. INTRODUCTION

Covid-19 pandemic introduced us to the world of quarantining and working from home. It also brought a significant amount of changes in the field of education. Online and blended modes of education have received greater acceptance by the academic community [1], [2]. The decision to continue with these modes of delivery in this post-COVID-19 pandemic era needs to be evaluated. Performance prediction of students' can help in evaluating the effectiveness of online and blended mode of delivery. Hence, the performance prediction of students' has gained a greater significance in academic society. Students' performance is influenced by the interaction of students with course content, peers, and the course instructors. The learning ecosystems established around the student are expected to ensure that these three interactions effectively engage the student. The assessment of student's performance is an important attribute of the learning ecosystem, as it provides feedback to the students as well as the instructors [3]. The predicted performance of the student can help instructors monitor the learning progress of the students and provide feedback. It also allows them to customise the interventions to meet the student's needs. This can further help students to improve their performance [3].

There is a plethora of research studies available on student performance prediction. Several previous research studies have found that the data collected from online learning platforms is useful in predicting student performance [4], [5]. On the other hand, student participation in social media platforms and discussion forums has also been used to predict student performance [5]–[8]. Studies also show that demographic characteristics influence students' performance [9]–[11]. Some works also propose the usage of multimodal data to understand students' behavior [12].

In the horizon of the aforementioned situation, the study is situated in the context of a private engineering university where a blended mode of delivery was adopted at university level [1]. The blended mode of delivery combined online and physical modes of learning. The courses were delivered through the institution's moodle-based learning platform, the Learning Management System (LMS). The content was delivered in asynchronous mode and synchronous mode. Studio-based videos, screencasting videos, demonstration videos, and reading materials were made available to the students as a part of the asynchronous mode of delivery [2]. The synchronous mode of delivery included hands-on activities such as group projects and formative assessments conducted to evaluate students. The university was in need of evaluating the effectiveness of this blended learning in terms of

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students' performance to decide whether to continue with it even after the pandemic. The described situation served as motivation for initiating the current research study.

Artificial intelligence (AI) has been growing extensively and has paved it's way into many fields including the field of education [13]. It has been researched and used to predict student performance by various researchers [10]. A systematic literature survey(SLR) [14] was conducted and the outcome of the SLR helped the authors to arrive at list of constructs and machine learning (ML) models used by researchers towards predicting the student performance. Some of the ML models used by the researchers include logistic regression [10], random forest [6], [9], naive Bayes [7], [15], decision tree [4], [11], support vector machine(SVM) [7], [9], [16], and XGBoost [16].

The advancement of ML and DL algorithms towards application in the field of education helped the authors to define the scope of this study. The data included in the study lies in the first-year engineering course titled engineering exploration [17] The reason behind the choice of the course for the study is the nature of the course itself. It is an Project Based Learning (PBL) course [18] and the course demands the interaction of students with all three dimensions including content, peer and instructor. The study was initiated with the following research questions:

1) How the current practices of establishing interaction of student with the content, peers, and instructors are influencing the student's performance?
2) How does the demography (gender) of the student influence the performance in the online and blended mode of delivery of a first-year engineering course?

Students with diverse backgrounds [19], learning styles [20], and varied motivation levels enroll for engineering degrees. Some students enroll in engineering degrees due to peer and/or parental pressure and as a result lack motivation. Early failure prediction of such students help the instructors make timely interventions and scaffold students.

The study involves investigation of student interactions in three dimensions including interaction with the content, interaction with the peers, and interaction with the teachers. The collection of data to understand the three dimensions resulted in heterogeneous data that needs to be fed to the ML model. The modalities of data considered in the study included click stream data collected through LMS that reflected the student's interaction with the content, GitHub contributions, and collaborations that reflected the interactions with the peer acted as another modality while demographic details was the other one.

The rest of the paper is organised as follows. Related work is reviewed in Section II. The methodology adopted is described in Section III. The results are discussed in Section IV, followed by inferences and future directions in the conclusion section.

II. RELATED WORKS

Student performance prediction has been studied extensively in the past for various purposes like identifying the students at risk of failing and identifying and customising the student learning environments. The predicted values are the grade point averages, knowledge, and scores [21].Generally, the body of work used classification approaches to classify students into discrete categories such as "Pass" or "Fail" [9], [11]. Most works have used quantitative approaches. Some works refer to the context where qualitative analysis [22] is performed. In the study [22], 27 students from an African classroom were considered and their interaction with content was qualitatively analyzed. The authors of [22] concluded that there is a strong correlation between performance and engagement implying the students who were more engaged would perform better than those less engaged.

In [5], the authors used both qualitative and quantitative methods. They used statistical information like the number of messages, posts read, times visited, etc. for the quantitative analysis and performed qualitative analysis to understand the content of messages. They concluded that the combination of qualitative and quantitative measures improves the prediction accuracy with a value of 90.3%.

In [10] the authors used a combination of demographic factors, student's past academic records, and clickstream data of that students which was collected from their activities in the virtual learning environment. They defined the students with a predicted performance below 40% as students at risk of failing. They have concluded that demographic characteristics and students' clickstream activities have a significant impact on student performance. Artificial neural networks (ANNs) give better results with an accuracy of 94% when compared to baseline classification models like support vector machines (SVM) and logistic regression(LR).

Peer interaction is also an important factor influencing students' performance. In the work [6], the authors have used the activities of students on social media platforms like wiki, blog posts, and Twitter, to predict the students' performance. The authors considered factors like the number of posts a student posted on the platform, how many times the student was active on a particular day, and the length of their posts. They concluded that students' actions on social media tools are good predictors of academic performance. They also concluded that the large margin nearest neighbor regression (LMNNR) algorithm proved very suitable for the task of prediction achieving an accuracy of 85%.

Previous academic records of the student also can be used in predicting the student's performance [23]. In this study, the authors propose the usage of the previous internal assessment marks of the students to predict whether the student will fail or pass the final exam. They experimented with Adaptive Boosting, Deep Neural Networks(DNN), and artificial immune recognition system v2.0 for the task of classification. Their results indicate that DNN performed the best with an accuracy of 95.34%.

The factors discovered in the literature can be divided into four categories: demographic characteristics, academic records, content interaction, and peer interaction

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[5]. A complete discussion of these factors may be found in Table I. The algorithms utilized were divided into two categories: machine learning (ML) and deep learning(DL). Table II lists the various ML and DL algorithms that were used in this study. In general, classification approaches are seen to be employed more frequently.

The student's performance was described using a variety of criteria. Using multi-class classification techniques, they were typically categorized as pass/fail/distinction or other similar categories. These classifications were determined based on the final results of the tests (grade point averages or the total final marks) [9], [11].

TABLE I: Attributes Commonly Used

Identified Category	Attributes Considered	References
Demographic Factors	Gender, age, parent's educational level, parents' involvement, students study environment, school and classroom environment, possession of computers, region, family income and expenditure, nationality, birthplace.	[7], [9]–[11]
Academic records	Internal assessment marks, assignment submission marks, past performances.	[23]
Content interaction	Click stream data, number of clicks per day, number of active days, number of resources accessed, number of videos watched, announcement views.	[4], [7], [9], [10], [22]
Peer interaction	Activities of students on wiki, blog posts and twitter or any other discussion forums.	[6]–[8]

TABLE II: Methods Used in this Study

Methods	Approaches	References
Machine Learning	Logistic Regression, SVM, Naive Bayes, Decision Tree, Random Forest , XGBboost.	[4], [6], [7], [11]
Deep Learning	Deep Neural Networks(DNN), Long short term memory(LSTM),etc.	[10], [22]

TABLE III: Factors Used and Their Descriptions

Factors	Description
Interaction with Content (Click Stream Data)	Total number of times the student visited LMS
	Number of times the student visited the website per day on an average
Interaction with Peers	Total number of Commits made by each student on GitHub.
Academic Performances	Formative assessment marks (I 1-3 marks of 8 consecutive assessments conducted in the physical mode of delivery)
Demographic Factors	Gender

III. METHODOLOGY

For the purpose of this study, 140 students pursuing the course of engineering exploration [1], [17] were considered. The course uses project-based learning [18] pedagogy. Students work in teams to develop a functional prototype as a solution towards the identified problem [24]–[26]. The course focuses on first-year engineering students and teaches them about the engineering design process, the multidisciplinary nature of engineering, problem-solving, data analysis, team building, professional ethics, sustainability, and project management [17]. Students use agile practice [27] to complete the project. This course was delivered using the blended approach for the semester Jan-Jun 2020 [1]. This transition from physical mode to online mode opened up new research studies in the direction of measuring the effectiveness of blended learning and personalized learning. The factors and methods identified in the previous section were considered while building the implementation pipeline.

A. Data Collection

The data used consisted of 140 students pursuing the course engineering exploration. The online mode and physical mode of learning constitute the blended mode (synchronous and asynchronous) of delivery. Clickstream data was the reflection of students' interaction with the content, and GitHub collaboration was the reflection of students' interaction with their peers. These two factors were considered under the asynchronous mode of delivery. The previous performance (marks of the previous formative assessment) of the student is considered under the synchronous mode. The marks of the formative assessments conducted from January to March 2020 were used to predict the performance of students in formative assessments scheduled in the next part of the semester (after March) as well the semester-end exams.

Students' performance is influenced by their interaction with peers, content, and instructors [28]. LMS set up by the university is a moodle based online platform through which the content of the course was accessed along with the required resources for the course. The clickstream data, that was obtained from the LMS platform, was considered as a part of the interaction with the content dimension. The GitHub collaborations made by each student were considered under the interaction with peers dimension. Other than this the previous academic records of the students were considered along with gender of the student. The multimodal data, i.e., clickstream data, Github collaborations, academic records, and demographic data, collected from different sources were used to further this study. Figure 1, shows the implementation pipeline adopted to classify the students based on their performance. Table III, lists all the factors used for the purpose of this study in detail.

B. Data integration and Cleaning

Integration of the multimodal data was a crucial part of this study. Early data fusion techniques were used to arrive at the fused data. The missing values in the data were replaced with the mode.

C. Classifier

The processing of the data and integration of the data was done using Python libraries such as NumPy and Pandas. The cleaned and fused data was split into train and test data with a ratio of 80-20. The training data was used to develop five different machine learning models namely: random forest, naive Bayes, decision tree, support vector machine (SVM), and XGBoost. The parameters for these

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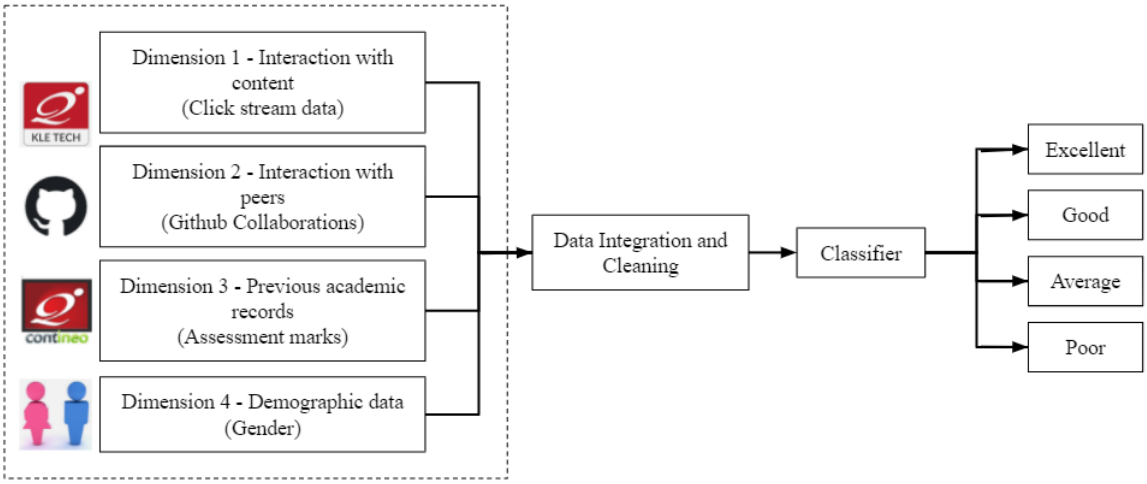

Fig. 1 Proposed pipeline for implementation

classifiers were set using grid search cross-validation.

D. Predicted Output

The output of the classification task is the prediction of students' performance. The students were graded out of 20. The classifiers decided the category to which the student belonged to based on the input data. The students were categorised into 4 categories. The categories were as follows:

- Excellent: Final score is greater than 15
- Good: Final score is greater than 10 and less than 15
- Average: Final score is greater than 5 and less than 10
- Poor: Final score is less than 5

The number of students present in each category "poor", "average", "good", and "excellent" was 29, 31, 43, 37 respectively. The evaluation metric considered in this work is the accuracy of the machine learning classifiers. The accuracy of a machine learning model is a metric for determining which model is the best at recognising relationships and patterns between variables in a dataset based on the input or the training data. The formula to calculate the accuracy is provided in equation 1.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$TP = True_{Positive}, TN = True_{Negative},$$
$$FP = False_{Positive}, FN = False_{Negative}$$

IV. RESULTS AND DISCUSSION

To address the mentioned research questions, data was collected and analysed. A correlation heatmap of all the parameters or factors considered in this work results is shown in Figure 2.

The correlation heatmap provides information regarding the amount of correlation between the factors considered. From the above matrix, it was observed that the total number of times a student visits the LMS platform has a positive correlation of 0.68 with the final grade of the student. That implies the more the number of visits, the higher will be the grade of the student. This in turn tells

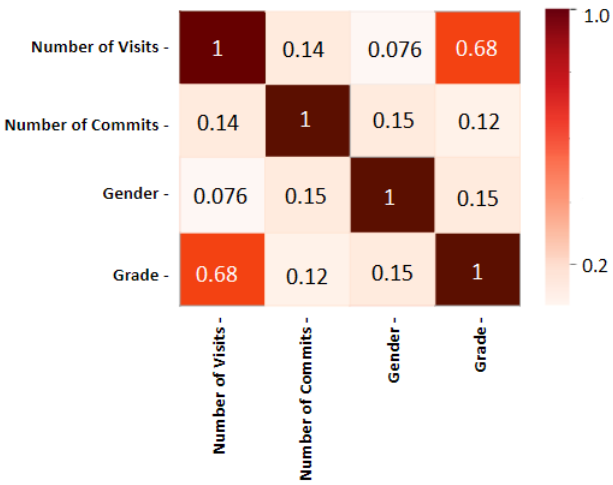

Fig. 2 Correlation heatmap

us that the interaction with content has a positive influence on the student's performance. The number of commits has a low positive correlation of 0.12. This suggests that the interaction of students with peers does not have a linear relationship with student's performance. The demographic data(gender) also does not have a linear relationship with students' performance. It has a correlation of 0.15 with the student's performance. The results obtained from all the classifiers can be observed in table IV:

TABLE IV: Comparison of the results obtained by the ML algorithms on the test data

Model	Accuracy
Decision tree	96%
Random forest	93%
SVM	93%
Naive Bayes	89%
XGBoost	79%

From table IV, we can observe that the decision tree, random forest, SVM, naive Bayes, and XGBoost gave an accuracy of 96%, 93%, 93%, 89%, and 79% respectively. Clearly, the decision tree classifier outperformed all the other classifiers with an accuracy of 96 %. Figure 3, represents the validation curve of the decision tree classifier.

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It shows that the model has low bias and low variance.

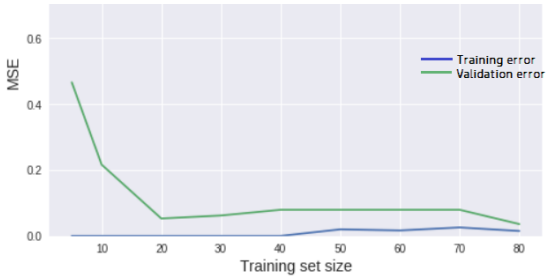

Fig. 3 Validation curve of decision tree classifier

V. CONCLUSION

The study shows that the current practices of establishing interaction of students with the content have a positive influence on the student's performance. The interaction with peers reflected through GitHub contributions, and demographic data (gender) do not have a linear relationship with the student's performance. The study also shows that the use of multiple modalities not only helped the triangulation but also in training the classifiers. The decision tree classifier performed best with an accuracy of 96%.

Studying the third dimension of interaction, i.e., interaction with instructors, and its influence on student performance is the future scope of this study. Along with this the interdependencies among the three dimensions of interaction also needs to be studied. The following study was limited to a set of 140 students from a particular institution. A study including a larger and more generalized dataset will ensure that the models also get trained in a generalized manner.

REFERENCES

[1] M. Vijaylakshmi, P. Baligar, K. Mallibhat, S. M. Kavale, G. Joshi, and A. Shettar, "Transition from in-person learning to technology enhanced learning in engineering education: Faculty challenges," pp. 1–5, 2021.

[2] K. Mallibhat and N. Iyer, "Program level adaptations during pandemic: Opportunities and faculty perceptions," pp. 3138–3145, 4-6 July, 2022 2022. [Online]. Available: https://dx.doi.org/10.21125/edulearn.2022.0778

[3] W. Xing, R. Guo, E. Petakovic, and S. Goggins, "Participation-based student final performance prediction model through interpretable genetic programming: Integrating learning analytics, educational data mining and theory," *Computers in Human Behavior*, vol. 47, pp. 168–181, 2015.

[4] M. Hussain, W. Zhu, W. Zhang, and S. M. R. Abidi, "Student engagement predictions in an e-learning system and their impact on student course assessment scores," *Computational intelligence and neuroscience*, vol. 2018, 2018.

[5] L. P. Macfadyen and S. Dawson, "Mining lms data to develop an "early warning system" for educators: A proof of concept," *Computers & education*, vol. 54, no. 2, pp. 588–599, 2010.

[6] E. Popescu and F. Leon, "Predicting academic performance based on learner traces in a social learning environment," *IEEE Access*, vol. 6, pp. 72 774–72 785, 2018.

[7] V. Vamshidharreddy, A. Saketh, and R. Gnanajeyaraman, "Student's academic performance prediction using machine learning approach," *Int. J. Adv. Sci. Technol*, vol. 29, no. 9, pp. 6731–6737, 2020.

[8] C. Romero, M.-I. López, J.-M. Luna, and S. Ventura, "Predicting students' final performance from participation in on-line discussion forums," *Computers & Education*, vol. 68, pp. 458–472, 2013.

[9] P. Sokkhey and T. Okazaki, "Hybrid machine learning algorithms for predicting academic performance," *Int. J. Adv. Comput. Sci. Appl*, vol. 11, no. 1, pp. 32–41, 2020.

[10] H. Waheed, S.-U. Hassan, N. R. Aljohani, J. Hardman, S. Alelyani, and R. Nawaz, "Predicting academic performance of students from vle big data using deep learning models," *Computers in Human behavior*, vol. 104, p. 106189, 2020.

[11] S. Rizvi, B. Rienties, and S. A. Khoja, "The role of demographics in online learning; a decision tree based approach," *Computers & Education*, vol. 137, pp. 32–47, 2019.

[12] M. N. Giannakos, K. Sharma, I. O. Pappas, V. Kostakos, and E. Velloso, "Multimodal data as a means to understand the learning experience," *International Journal of Information Management*, vol. 48, pp. 108–119, 2019.

[13] A. Hellas, P. Ihantola, A. Petersen, V. V. Ajanovski, M. Gutica, T. Hynninen, A. Knutas, J. Leinonen, C. Messom, and S. N. Liao, "Predicting academic performance: a systematic literature review," in *Proceedings companion of the 23rd annual ACM conference on innovation and technology in computer science education*, 2018, pp. 175–199.

[14] Y. Joshi, M. Kaushik, and M. Vijaylakshmi, "Systematic literature review: An investigation towards finding constructs for performance prediction of students in an online engineering course," *2022 World Engineering Education Forum and The Global Engineering Deans Council (WEEFGEDC)*, pp. 1–5, 2022.

[15] A. Daud, N. R. Aljohani, R. A. Abbasi, M. D. Lytras, F. Abbas, and J. S. Alowibdi, "Predicting student performance using advanced learning analytics," in *Proceedings of the 26th international conference on world wide web companion*, 2017, pp. 415–421.

[16] B. Sekeroglu, K. Dimililer, and K. Tuncal, "Student performance prediction and classification using machine learning algorithms," in *Proceedings of the 2019 8th International Conference on Educational and Information Technology*, 2019, pp. 7–11.

[17] P. Baligar, S. Kavale, M. Kaushik, G. Joshi, and A. Shettar, "Engineering exploration: A collaborative experience of designing and evolving a freshman course," pp. 1–5, 2018.

[18] M. Kaushik, "Evaluating a first-year engineering course for project based learning (pbl) essentials," *Procedia Computer Science*, vol. 172, pp. 364–369, 2020.

[19] N. N. Kulkarni, M. Kaushik, and G. Joshi, "Engineering profession: understanding freshman perspective," in *2016 IEEE 4th International Conference on MOOCs, Innovation and Technology in Education (MITE)*. IEEE, 2016, pp. 332–337.

[20] M. Kaushik and G. Joshi, "Transitional learning style preferences and its factors in newer generation engineering students," in *2016 IEEE 4th International Conference on MOOCs, Innovation and Technology in Education (MITE)*. IEEE, 2016, pp. 263–267.

[21] A. Hellas, P. Ihantola, A. Petersen, V. V. Ajanovski, M. Gutica, T. Hynninen, A. Knutas, J. Leinonen, C. Messom, and S. N. Liao, "Predicting academic performance: a systematic literature review," pp. 175–199, 2018.

[22] J. Mutahi, A. Kinai, N. Bore, A. Diriye, and K. Weldemariam, "Studying engagement and performance with learning technology in an african classroom," in *Proceedings of the Seventh International Learning Analytics & Knowledge Conference*, 2017, pp. 148–152.

[23] S. Hussain, Z. Muhsin, Y. Salal, P. Theodorou, F. Kurtoğlu, and G. Hazarika, "Prediction model on student performance based on internal assessment using deep learning," *International Journal of Emerging Technologies in Learning*, vol. 14, no. 8, 2019.

[24] M. Kaushik, P. Baligar, and G. Joshi, "Formulating an engineering design problem: A structured approach," *Journal of Engineering Education Transformations*, 2018.

[25] P. Baligar, K. Mallibhat, S. Kavale, and G. Joshi, "Evaluating an engineering design problem for its complexity," *IEEE Transactions on Education*, vol. 65, no. 1, pp. 73–80, 2021.

[26] M. Kaushik, P. Pillai, and N. C. Iyer, "Prototype implementation: An effective learning method in process automation," *Journal of Engineering Education Transformations*, vol. 28, no. 2 & 3, pp. 45–50, 2015.

[27] J. Gadad, V. Talageri, M. Kaushik, and M. Asundi, "An experience of implementing agile project management practice in pbl course at first-year engineering course," in *EDULEARN22 Proceedings*, ser. 14th International Conference on Education and New Learning Technologies. IATED, 4-6 July, 2022 2022, pp. 5391–5398. [Online]. Available: https://dx.doi.org/10.21125/edulearn.2022.1274

[28] T. Anderson, "Modes of interaction in distance education: Recent developments and research questions," *Handbook of distance education*, pp. 129–144, 2003.

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